Experiment: Study of LAN Topology, Network devices and Address Configuration.

Objective:→

① To study the LAN topology used in college campus.
② To identify networking device & understand their function.
③ To analyze wired & wireless communication method used in the network.
④ To measure speed of network for wired & wireless connections.
⑤ To obtained MAC, IP Address & Subnet mask.

Problem statement-1

• Draw & identify LAN topology of college campus & clearly label all components such as router, switches etc.

Soln: The college campus network observed this study consists of:→

* Layer-3 Core switches in the server Room.
* Managed by POE + switches.
* wireless Access point (APs)
* Ethernet port in classroom & Laboratories.

Question no:1

→ **Network Topology**

Hybrid Hierarchical Topology which combines:

1. Tree Topology.
2. Extended Star Topology
3. Ring Topology (Server room).

→ **The network hierarchical layer:→**

① Core layer.
② Distribution layer.
③ Access layer

① **Core layer:→** Core layer contains Server Room and server room have firewall, Router, 2-L3+ Switches, Patch Rack, Server, inverter etc.

② **Distribution layer:→**

* Distribution layer contains 4- Switches
* The 4 Switches are:→.

① Main Build → R7 (Room no-213, 2nd floor).
② Main Build → R1 (Room no.→ 301, 3rd floor).
③ Main Build → R6 (Room no-113, 1st floor).
④ Main Build → R2 (Room no. 201, 2nd floor).

* This 4- Switches contains:→

① 18 Port Gigabit Easy Smart Switch with 16- Port. TL-SG1218MPE

② SG3428XMP → 24 Port Gigabit & 4- port 10GE SFP+, L2+ Managed Swichtes switch. with 24 port POE+.

→ Now let's discuss these 2 above components :→

① TP-link TL-SG1218MPE

* Likely used for : CCTV cammera , wifi access point, etc.
* 18 port Gigabit "Easy Smart" Switch.
* 16 port Support POE+

② TP-link Omada SG3428XMP.

* Higher-end managed layer 2+ Switch.
* 24 POE+ Port , uplinks , 10G SFP for each floor.

→ Access Layer :→

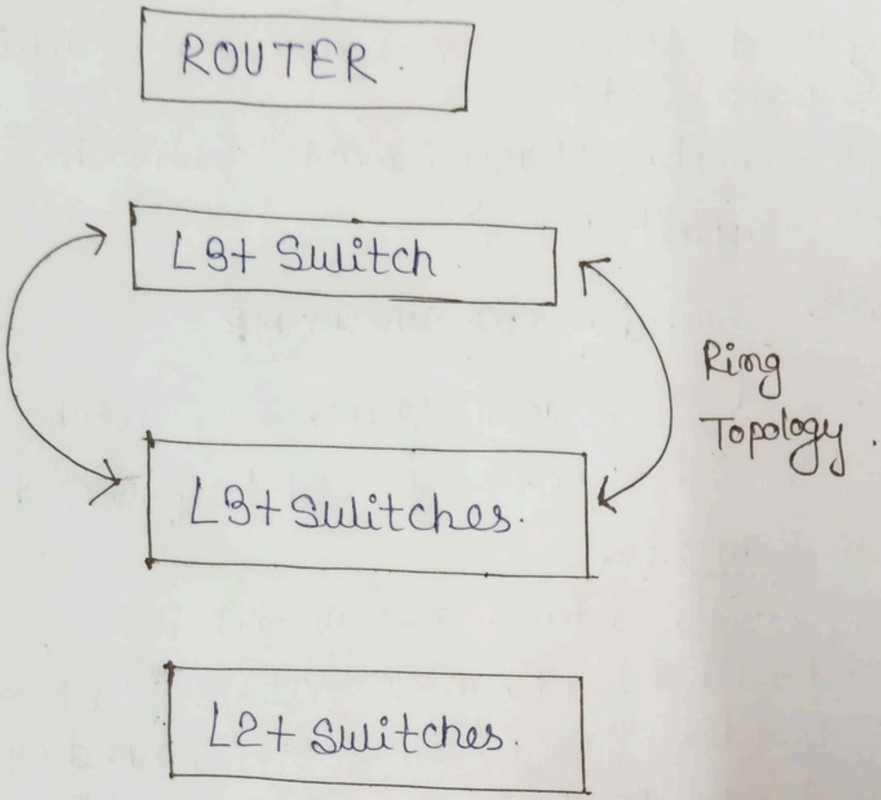
* Access Layer Contain 3-Switches i.e :→
(I) Main Build R1-[Room no-301 , 3rd floor].
(II) Main Build R4 [Room no-05 , Ground floor]
(III) Main Build R5 [Room no-18 , Ground floor].

* Every switches act as access Layer which helps to provide network connectivity to all classroom , Lab , faculty Room , etc.

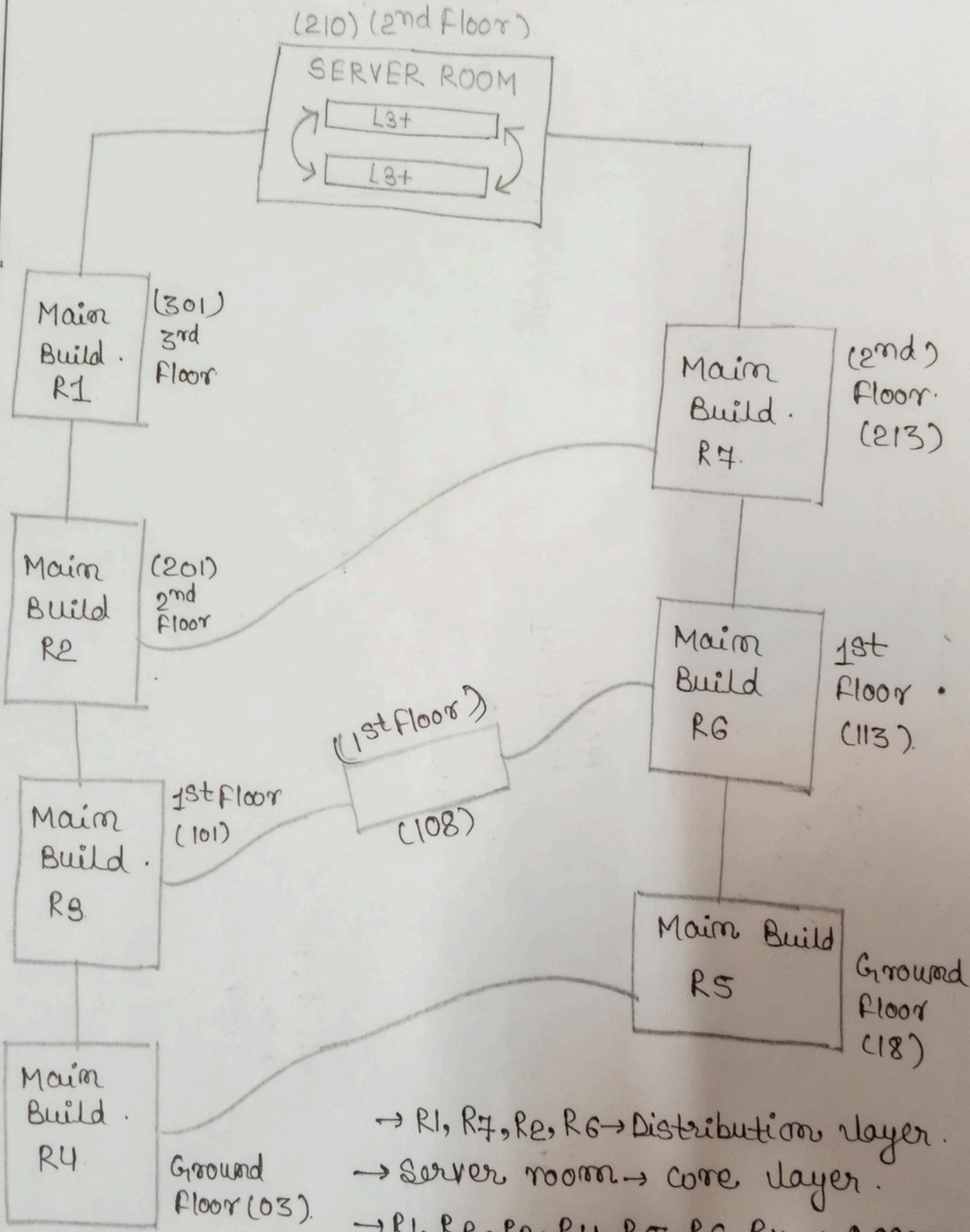
* Access Layer directly connect :→
① Ethernet Port
② Wireless AP.
③ Lab System.

④

→ Server Room Architecture/Topology :→



TOPOLOGY

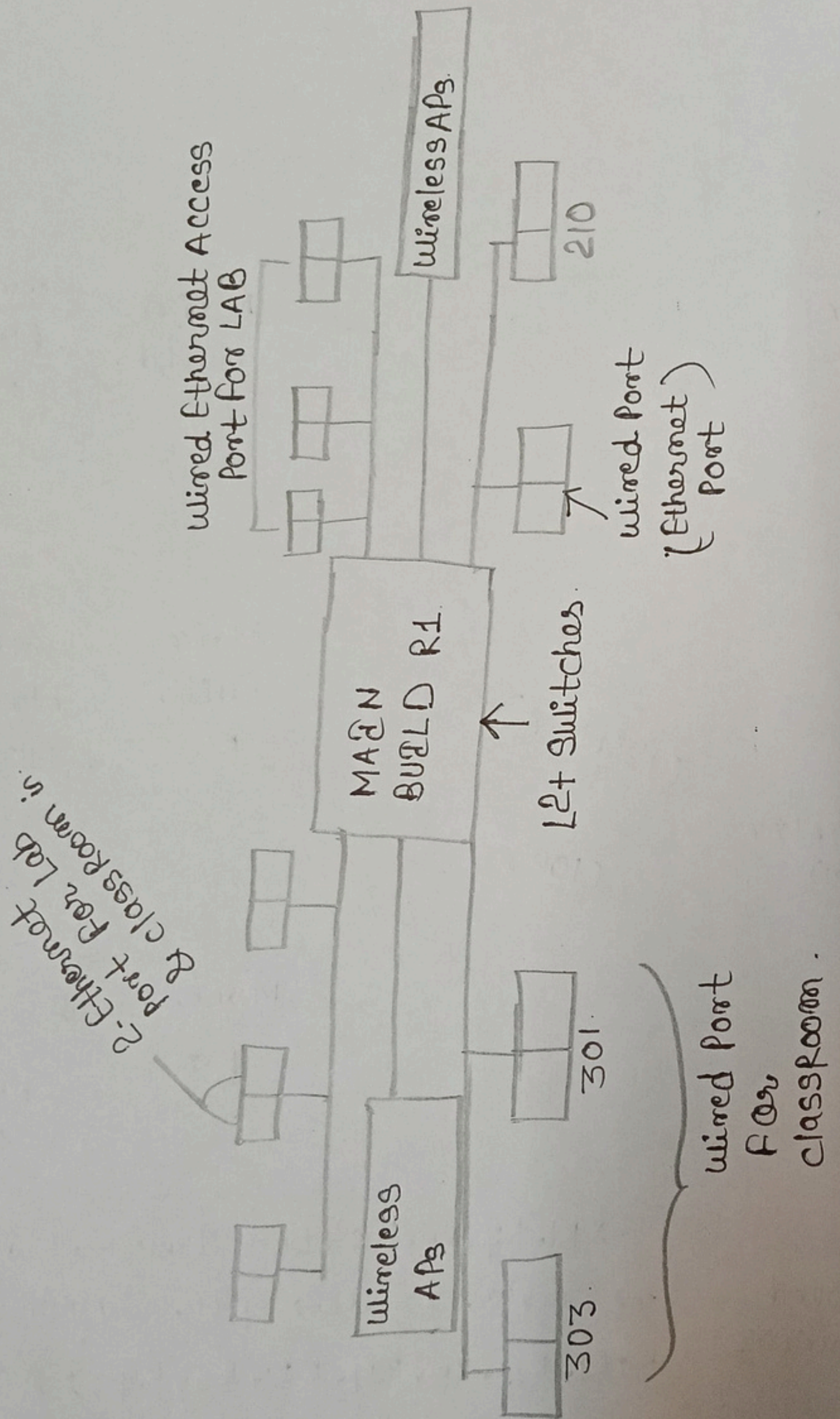


→ R1, R7, R2, R6 → Distribution layer.

→ Server room → core layer.

→ R1, R2, R3, R4, R5, R6, R7 → Access layer.

ACCESS LAYER SWITCH (3rd Floor 301)



2. Identify Network Devices :→
- list all networking devices used in the LAN
- Mention the function of each device.

1. Router

Function:→

- Connects the LAN to the Internet or other networks.
- Assigns IP addresses (DHCP).
- Routes data packets between different network.

2. Switch

Function:→

- Connects multiple devices (PCs, printers, etc) within the LAN.
- Transfers data to the Correct device using MAC addresses.

3. Wireless Access Point / Wi-Fi Router

Function:→

- Provides wireless (Wi-Fi) Connectivity to devices in the LAN.
- Allows laptops/mobile devices to join the Network.

4. Network Interface Card (NIC) / Network Adapter

Function:→

- Hardware inside a computer that enables network connection.
- Example: Wi-Fi Adapter, VM ware Network Adapter.

5. Computer/HOST Device

Function:→

- End device used to Send, receive, and process data on the LAN.

Question no:2

6. Ethernet Cable.
(if Wired LAN is used)

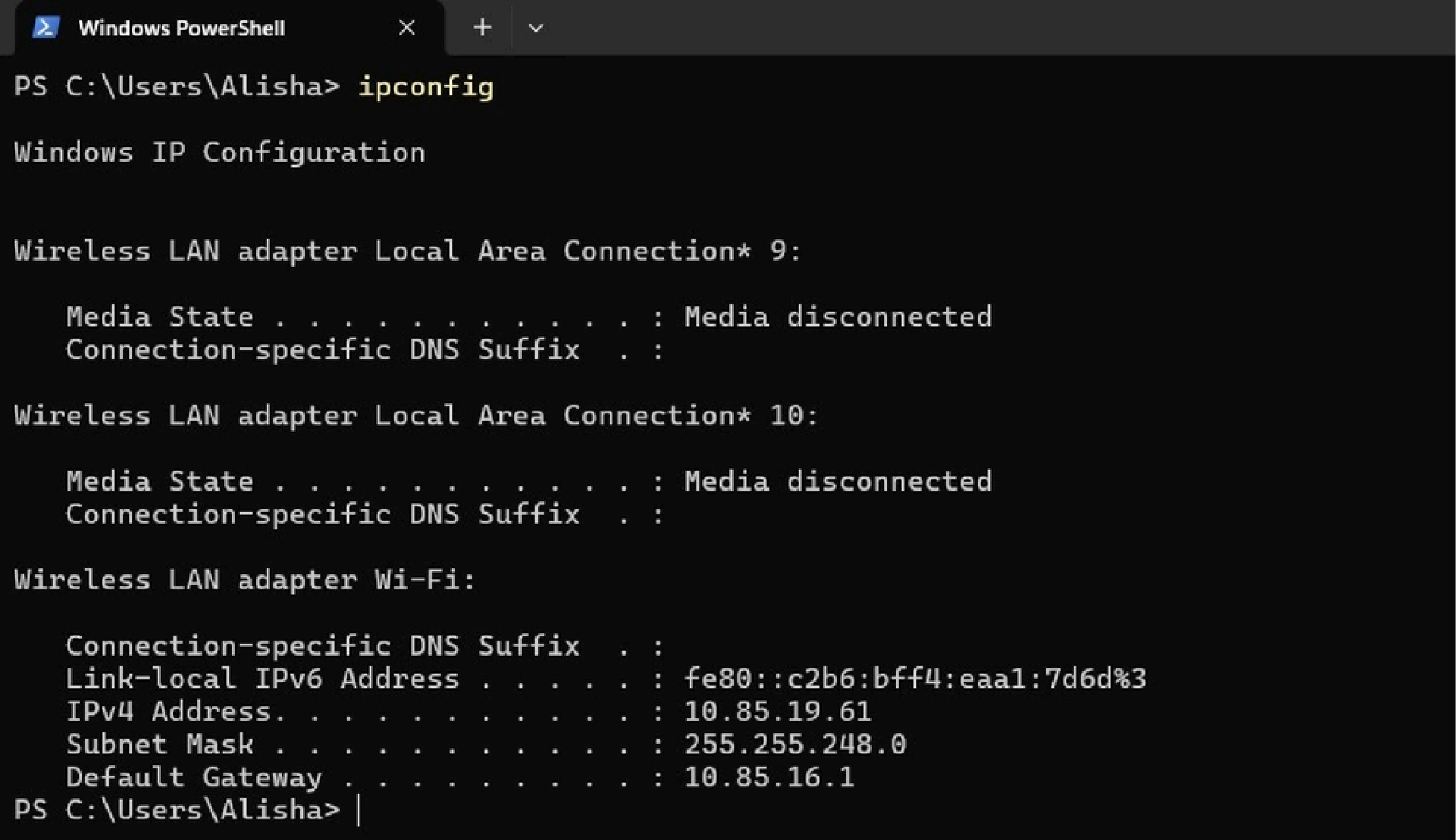
Function :→

- Physically connects devices like Computers, Switches and routers for data transfer.

7. Modem
(if Internet access is included)

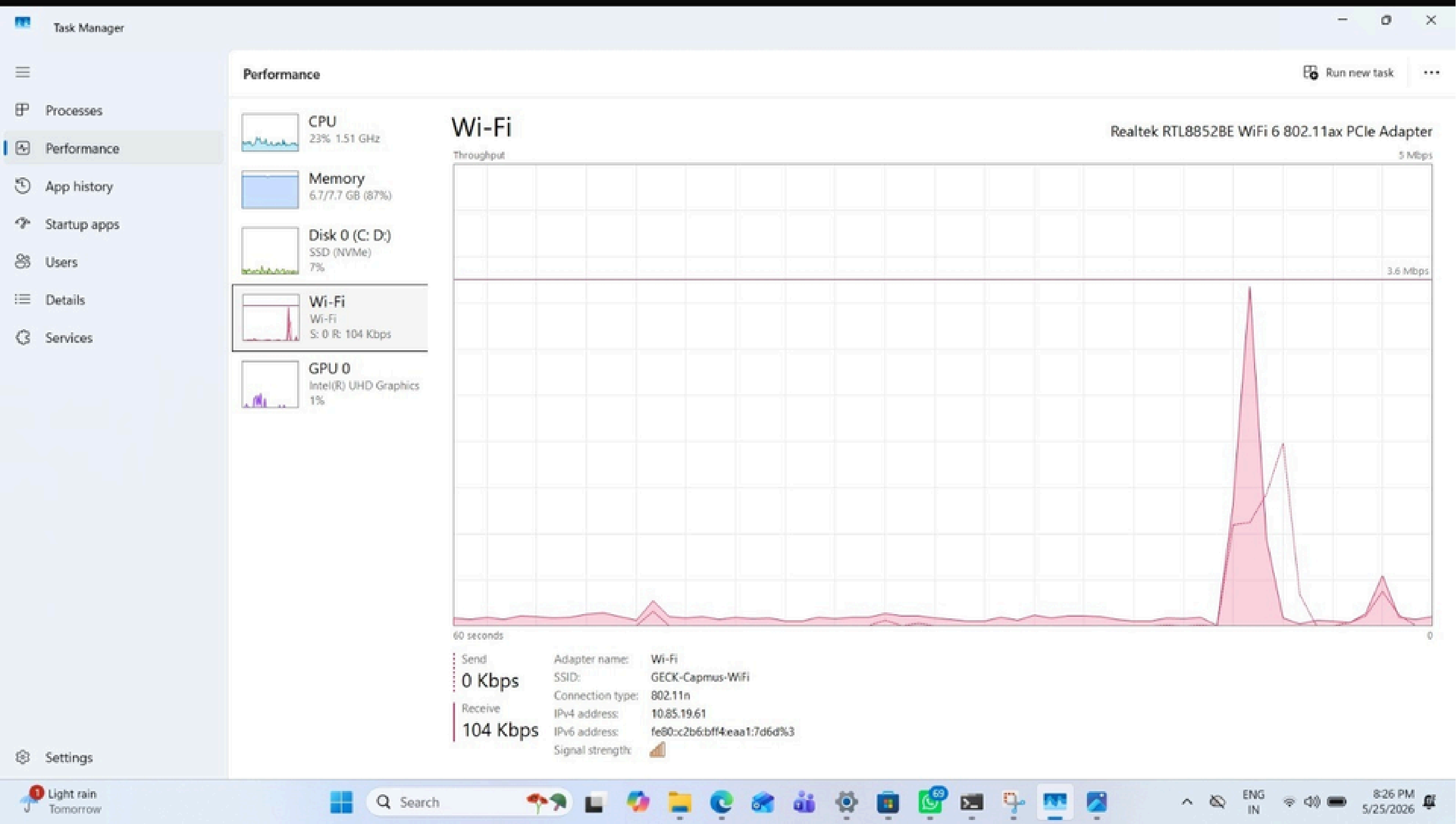
Function :→

- Connects the router/LAN to the ISP (Internet Services Provider)
- Converts ISP signals into digital network signals

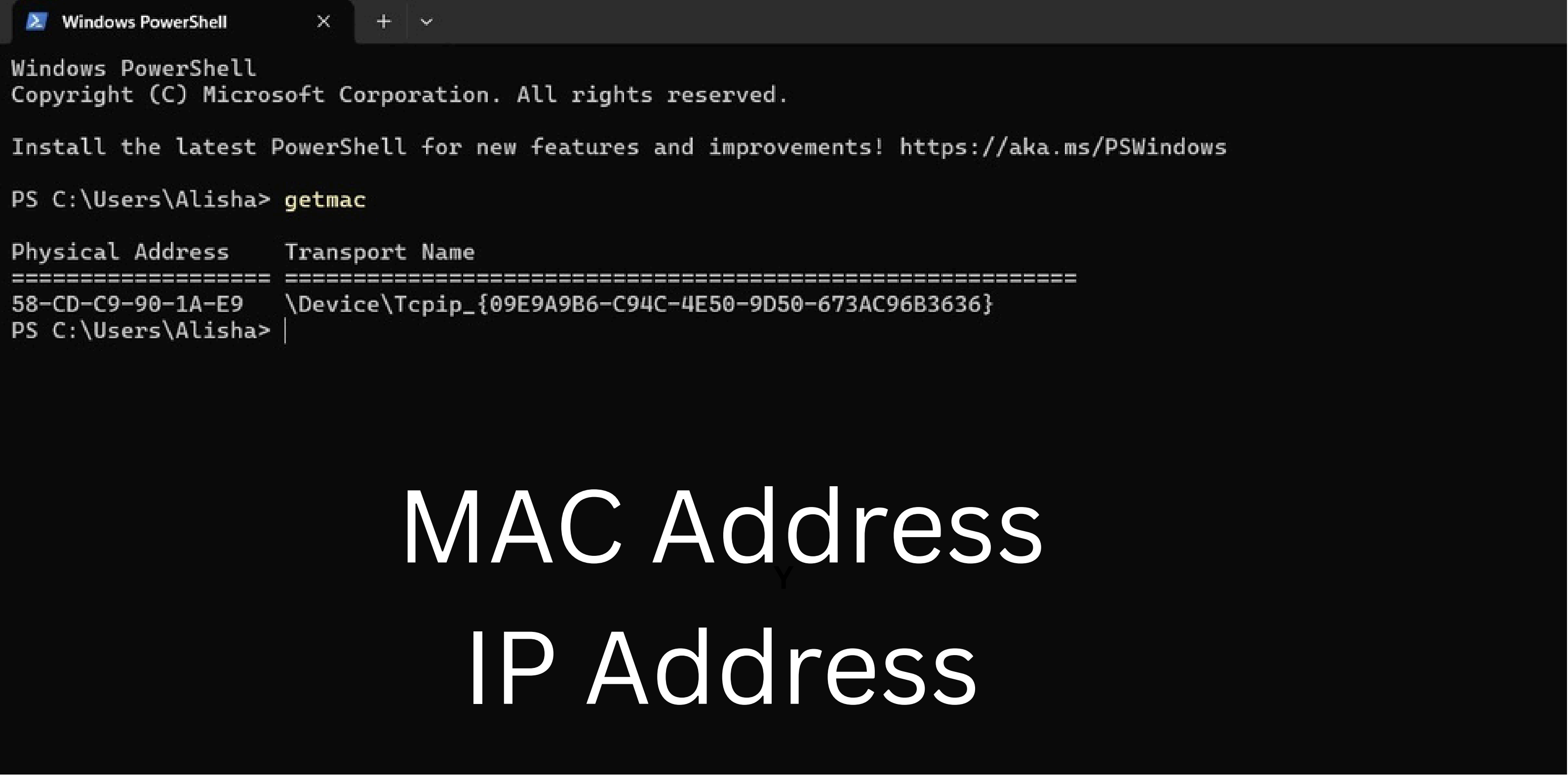


```
PS C:\Users\Alisha> ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c2b6:bff4:eaa1:7d6d%3
   IPv4 Address. . . . . . . . . . . : 10.85.19.61
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 10.85.16.1
PS C:\Users\Alisha> |
```

Wireless Connection



Network Speed(104Kbps)


Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\Alisha> getmac
Physical Address   Transport Name
=================== ==
58-CD-C9-90-1A-E9   \Device\Tcpip_{09E9A9B6-C94C-4E50-9D50-673AC96B3636}
PS C:\Users\Alisha>
MAC Address
IP Address

6. Explain Observations:

- Briefly explain:
  - → Difference between wiered and wireless speed.
  - → Role of subnet mask in networking.

1. ans:

Wired Network	Wireless Network
• Uses Ethernet Cable for connection.	• Uses Wi-Fi signals for Connection.
• Generally faster and more stable.	• Speed may vary due to distance and interence.
• Lower latency (faster response time).	• Higher latency compared to wired networks.
• More secure because physical access is required.	• Less secure if Wi-Fi security is weak.

Example: In a computer lab, a PC connected through an Ethernet Cable may get 100Mbps, while a laptop connected through Wi-Fi may get 60-80 Mbps depending on signal strength.

2. Role of Subnet mask in networking: →

ans: A Subnet Mask is used to Identify the network portion and host portion of an IP address. It helps device determine whether another device is on the same network or a different network.

Example: • IP address: 192.168.1.10
• Subnet Mask: 255.255.255.0

Here: → • Network ID = 192.168.1.0
• HOST ID = 10

Question no:6

• All devices with addresses 192.168.1.x belong to the same network and can communicate directly

List of Devices and Their Roles :→

Device		Role
1. Router	:	Connects local Network to the Internet.
2. Switch	:	Connects Multiple devices with a LAN.
3. Computer/Laptop	:	Sends and receives data.
4. Access Point/wifi Router	:	Provides wireless connectivity
5. Modem	:	Connects Network to ISP.

Table Of Network Parameters :→

Parameter	Example Value
MAC Address.	00:1A:2B:3C:4D:5E
IP Address.	192.168.1.10
Default Gateway	192.168.1.1

Network Diagram :→

Internet
↓
Router
↓
Switch

PC-1 PC-2 Laptop